QUESTION 13: WORD FREQUENCY DISTRIBUTION FROM A TEXT FILE

13. Scenario: You are working on a text analysis project and need to determine the frequency distribution of words in a given text document. You have a text document named "sample_text.txt"containing a paragraph of text. Your task is to develop a Python program that reads the textdocument, processes the text, and generates a frequency distribution of the words.

Question: How would you develop a Python program to calculate the frequency distribution of words in a text document?

Aim: To calculate the frequency distribution of words in a text document using Python.

Algorithm:

1. Open the text file.
2. Read the text and convert it to lowercase.
3. Split the text into words.
4. Count each word using Counter.
5. Display the word frequencies.

Code:

```
from collections import Counter

with open("sample_text.txt", "r") as file:
    text = file.read().lower()

words = text.split()

frequency = Counter(words)

print("Word Frequency:")

print(frequency)
```

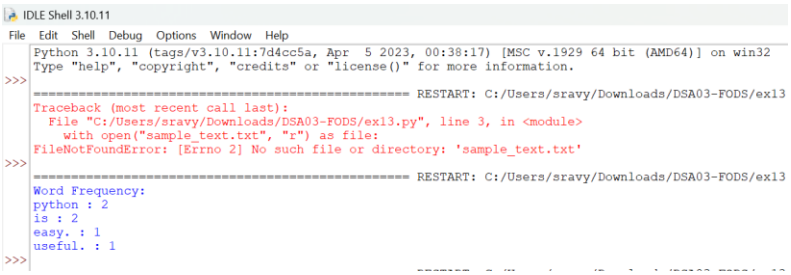
Input (sample_text.txt):

Python is easy. Python is useful.

Output:

Word Frequency:

Counter({'python': 2, 'is': 2, 'easy.': 1, 'useful.': 1})



Result: The frequency distribution of words was calculated successfully.